

Aggregation in SQL

Q1: Count how many cities are there in each country?

```
> SELECT CountryCode, COUNT(*) AS CityCount  
FROM city  
GROUP BY CountryCode;
```

Q2: Display all continents having more than 30 countries.

```
> SELECT Continent, COUNT(*) AS CountryCount  
FROM country  
GROUP BY Continent  
HAVING CountryCount > 30;
```

Q3: List regions whose total population exceeds 200 million.

```
> SELECT Region, SUM(Population) AS TotalPopulation  
FROM country  
GROUP BY Region  
HAVING TotalPopulation > 200000000;
```

Q4: Find the top 5 continents by average GNP per country.

```
> SELECT Continent, AVG(GNP) AS AvgGNP  
FROM country  
GROUP BY Continent  
ORDER BY AvgGNP DESC  
LIMIT 5;
```

Q5: Find the total number of official languages spoken in each continent.

```
> SELECT co.Continent, COUNT(cl.Language) AS  
OfficialLanguageCount  
FROM country co  
JOIN countrylanguage cl ON co.Code = cl.CountryCode  
WHERE cl.IsOfficial = 'T'  
GROUP BY co.Continent;
```

Q6: Find the maximum and minimum GNP for each continent.

```
> SELECT Continent, MAX(GNP) AS MaxGNP, MIN(GNP) AS  
MinGNP  
FROM country  
GROUP BY Continent;
```

Q7: Find the country with the highest average city population.

```
> SELECT co.Name, AVG(ci.Population) AS AvgCityPop  
FROM country co  
JOIN city ci ON co.Code = ci.CountryCode  
GROUP BY co.Name  
ORDER BY AvgCityPop DESC  
LIMIT 1;
```

Q8: List continents where the average city population is greater than 200,000.

```
> SELECT co.Continent, AVG(ci.Population) AS AvgCityPop  
FROM country co  
JOIN city ci ON co.Code = ci.CountryCode
```

GROUP BY co.Continent
HAVING AvgCityPop > 200000;

Q9: Find the total population and average life expectancy for each continent, ordered by average life expectancy descending.

```
> SELECT Continent, SUM(Population) AS TotalPop, AVG(LifeExpectancy)
AS AvgLifeExp
FROM country
GROUP BY Continent
ORDER BY AvgLifeExp DESC;
```

Q10: Find the top 3 continents with the highest average life expectancy, but only include those where the total population is over 200 million.

```
> SELECT Continent, AVG(LifeExpectancy) AS AvgLifeExp
FROM country
GROUP BY Continent
HAVING SUM(Population) > 200000000
ORDER BY AvgLifeExp DESC
LIMIT 3;
```